SMARTBRIDGE

Data Analytics with Tableau

PROJECT DOCUMENTATION

Heart Disease Analysis

Using Tableau for Data Visualization & Business Intelligence

Program	Data Analytics with Tableau
Category	Healthcare | Group Project
Institution	Smart Bridge Skill Wallet
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Table of Contents

Table of Contents 2
1.1 Project Overview 4
1.2 Business Problem Statement 4
1.3 Business Requirements 4
1.4 Real-World Use Case Scenarios 4
Scenario 1 — Clinical Decision Support (Dr. Sharma) 5
Scenario 2 — Public Health Policy (Ramesh, Government Health Dept.) 5
Scenario 3 — Patient Self-Monitoring (Anita, 45-year-old professional) 5
1.5 Social & Business Impact 5
2.1 Background & Prior Research 6
2.2 Key Findings from Existing Literature 6
2.3 Role of Data Visualization in Healthcare 6
2.4 Gap Addressed by This Project 6
3.1 Dataset Overview 7
3.2 Dataset Columns & Description 7
3.3 Key Dataset Statistics 8
3.4 Data Storage in Database 8
3.5 SQL Operations Performed ..... **Error! Bookmark not defined.**
3.6 Connecting Database to Tableau 8
4.1 Data Quality Assessment 9
4.2 Data Transformations in Tableau 9
4.2.1 Age Category Grouping 9
4.2.2 Calculated Field: HD Rate 9
4.2.3 Calculated Field: BMI Category 9
4.3 Data Filters Applied 10
4.4 Data Preparation for Story 10
5.1 Visualization Summary 11
5.2 Detailed Visualization Descriptions 12
5.2.1 Gender vs Heart Disease (Bar Chart) 12
5.2.2 Age vs Heart Disease (Grouped Bar Chart) 12
5.2.3 Diabetic vs Stroke (Bubble/Scatter Chart) 12
5.2.4 Impact of Smoking and Alcohol on Heart Disease 12
5.2.5 Stroke vs Heart Disease (Multi-Panel) 12
5.2.6 Racewise Heart Disease (Pie Chart) 12
5.2.7 General Health vs Heart Disease (Bubble Chart) 12

5.2.8 Physical Activity vs Heart Disease (Donut Chart) 12
5.2.9 Age & BMI vs Diabetic (Treemap) 13
5.2.10 People Got Stroke — Diabetes, Heart Disease & Age 13
6.1 Dashboard Design Principles 14
6.2 Dashboard 1 — Demographic & Lifestyle Overview 14
6.3 Dashboard 2 — Clinical & Activity Analysis 14
6.4 Dashboard 3 — BMI, Diabetes & Compound Risk 14
6.5 Tableau Public URL 15
7.1 Story Overview 16
7.2 Story Scenes 16
8.1 Performance Test Results 17
8.2 Performance Observations 17
9.1 Technology Stack 18
9.2 Website Pages 18
9.3 Tableau Embed Code Approach 18
9.4 Flask API Endpoints 19
10.1 Summary of Results 20
10.2 Key Analytical Findings 20
Finding 1: Age is the Dominant Risk Factor 20
Finding 2: Diabetes Significantly Amplifies Risk 20
Finding 3: Physical Inactivity is a Strong Predictor 20
Finding 4: BMI and Obesity Drive Early-Onset Risk 20
Finding 5: Smoking Compounds, But Isn't Dominant Alone 20
Finding 6: Self-Reported Health Correlates with Diagnosis 21
10.3 Recommendations 21
11.1 Step-by-Step Project Development Log 22
11.2 Team Contributions 23
11.3 Conclusion 23
11.4 Future Enhancements 24
11.5 References 24

01

INTRODUCTION & PROBLEM UNDERSTANDING

Defining the healthcare problem, business requirements, and project objectives.

1.1 Project Overview

Heart disease remains one of the leading causes of mortality worldwide, and its impact continues to grow due to lifestyle changes, poor dietary habits, and lack of physical activity. Despite advances in medicine, prevention and early detection remain critical in reducing the risk of severe outcomes.

This project leverages Tableau as a powerful data visualization and business intelligence tool to analyze a heart disease dataset. The goal is to transform raw health data into meaningful, interactive dashboards that highlight key risk factors, identify correlations, and support evidence-based decision-making for healthcare providers, policymakers, and individuals.

1.2 Business Problem Statement

Problem Statement

Analyzing large-scale health data related to heart disease — encompassing patient demographics, medical history, lifestyle choices, and clinical indicators — requires advanced tools for extracting actionable insights.

Healthcare professionals and policymakers lack an integrated visual platform to simultaneously analyze multiple risk factors and their interplay in heart disease development.

This project addresses that gap by building interactive Tableau dashboards that make complex health data accessible and interpretable for non-technical stakeholders.

1.3 Business Requirements

- Analyze patient demographics (age, gender, race) in relation to heart disease prevalence
- Visualize lifestyle factors (smoking, alcohol, physical activity) and their correlation with heart disease
- Identify clinical comorbidities (diabetes, stroke, asthma, kidney disease) linked to heart disease
- Create interactive, filterable dashboards for real-time exploration by healthcare professionals
- Embed dashboards into a web application (Flask) for broader accessibility
- Generate a Tableau Story that narrates the complete analytical journey

1.4 Real-World Use Case Scenarios

Scenario 1 — Clinical Decision Support (Dr. Sharma)

Dr. Sharma is a senior cardiologist at a metropolitan hospital. She wants to understand which lifestyle factors contribute most to heart disease among middle-aged patients. Using the Tableau dashboards, she can analyze patient data segmented by age, gender, BMI, cholesterol levels, and smoking habits. This helps her identify high-risk groups and design targeted awareness campaigns such as advising patients on weight management and smoking cessation.

Scenario 2 — Public Health Policy (Ramesh, Government Health Dept.)

Ramesh works with a government health department and is tasked with developing preventive health policies. Using the dashboards, he studies heart disease prevalence trends across different demographics, comparing lifestyle indicators and disease rates. He uses the findings to recommend policies: fitness programs in workplaces, stricter tobacco regulations, and subsidies for healthier food options.

Scenario 3 — Patient Self-Monitoring (Anita, 45-year-old professional)

Anita has a family history of heart disease and wants to proactively monitor her health risks. Simplified dashboards provided by her healthcare provider allow her to visualize risk factors such as cholesterol levels, blood pressure, and lifestyle habits compared to healthy benchmarks. The dashboard highlights actionable steps like increasing physical activity or reducing fat intake.

1.5 Social & Business Impact

- Early identification of at-risk populations can reduce hospitalizations and associated healthcare costs
- Data-driven policies can lead to systemic health improvements at the community level
- Empowering patients with visual health data promotes proactive, preventive healthcare behavior
- Supports healthcare providers in personalizing treatment plans based on multi-factor risk profiles

02

LITERATURE SURVEY

Review of existing research on heart disease risk factors and data visualization approaches.

2.1 Background & Prior Research

Cardiovascular disease (CVD) is the leading global cause of death, accounting for approximately 17.9 million deaths per year according to the World Health Organization (WHO). Research has consistently identified key modifiable risk factors including smoking, physical inactivity, obesity (high BMI), diabetes mellitus, and hypertension.

2.2 Key Findings from Existing Literature

- Age and Gender: Heart disease risk increases significantly after age 45 in men and after age 55 in women. Male patients show higher early-onset prevalence.
- BMI and Obesity: A BMI above 30 (obesity) correlates with a 2–3x higher risk of heart disease. Diabetic patients with high BMI form a particularly high-risk group.
- Smoking: Smokers have a 2–4x higher risk of heart disease compared to non-smokers. Risk reduction begins within 1 year of cessation.
- Diabetes: Diabetic patients are 2x more likely to develop heart disease. The relationship between diabetes, stroke, and heart disease is well-established in literature.
- Physical Activity: Regular physical activity reduces heart disease risk by 35%. Sedentary behavior is an independent risk factor even after controlling for BMI.
- Race and Socioeconomics: Black Americans experience higher rates of hypertension and heart disease compared to other racial groups, partly attributable to systemic health disparities.
- General Health Self-Assessment: Studies show that patients who rate their health as 'Poor' or 'Fair' have significantly higher rates of diagnosed cardiovascular conditions.

2.3 Role of Data Visualization in Healthcare

Business Intelligence (BI) tools like Tableau have been increasingly adopted in healthcare analytics. Research by Tableau Healthcare Institute (2023) demonstrates that visual dashboards reduce the time for clinical insight generation by up to 70% compared to traditional reporting. Interactive filters allow clinicians to segment populations dynamically, enabling personalized risk assessment that static reports cannot provide.

2.4 Gap Addressed by This Project

While individual studies analyze specific risk factors in isolation, this project integrates multiple simultaneous variables — demographics, lifestyle, clinical comorbidities — into a unified visual platform. The web-integrated Tableau Story format makes findings accessible to non-technical healthcare administrators and policy makers who may not have direct Tableau access.

03

DATA COLLECTION & EXTRACTION

Dataset description, database setup, SQL operations, and Tableau connection.

3.1 Dataset Overview

The dataset used in this project is the Heart Disease Health Indicators Dataset, derived from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) survey data. The version used contains 4,500 patient records and 18 feature columns covering demographics, lifestyle habits, and clinical health indicators.

3.2 Dataset Columns & Description

Column Name	Data Type	Description
HeartDisease	*Binary (Yes/No)*	Target variable — whether patient has heart disease
BMI	*Numeric (Float)*	Body Mass Index — weight divided by height squared
Smoking	*Binary (Yes/No)*	Smoked at least 100 cigarettes in lifetime
AlcoholDrinking	*Binary (Yes/No)*	Heavy drinker (14+/week men, 7+/week women)
Stroke	*Binary (Yes/No)*	History of stroke events
PhysicalHealth	*Integer (0–30)*	Days physical health was not good in past 30 days
MentalHealth	*Integer (0–30)*	Days mental health was not good in past 30 days
DiffWalking	*Binary (Yes/No)*	Difficulty walking or climbing stairs
Sex	*Categorical*	Biological sex: Male or Female
AgeCategory	*Categorical (Ordinal)*	Age bracket in 5-year intervals (18–24 to 80+)
Race	*Categorical*	Ethnicity: White, Black, Hispanic, Asian, Other, etc.
Diabetic	*Categorical*	No / Yes / Borderline / Yes (during pregnancy)
PhysicalActivity	*Binary (Yes/No)*	Physical activity outside work in past 30 days
GenHealth	*Ordinal (5-pt scale)*	Self-reported health: Excellent / Very Good / Good / Fair / Poor
SleepTime	*Numeric (Float)*	Average sleep hours per 24-hour period
Asthma	*Binary (Yes/No)*	History of asthma diagnosis
KidneyDisease	*Binary (Yes/No)*	History of kidney disease (excluding stones)
SkinCancer	*Binary (Yes/No)*	History of skin cancer diagnosis

3.3 Key Dataset Statistics

Statistic	Value
Total Records	4,500
Heart Disease Positive	483 (10.7%)
Female Patients (No HD)	2,377
Female Patients (Yes HD)	227
Male Patients (No HD)	1,640
Male Patients (Yes HD)	256
Smokers in Dataset	~1,800 (40%)
Diabetic Patients	~900 (20%)
Stroke History	~220 (4.9%)
Physically Active	~3,000 (66.7%)

3.4 Data Storage in Database

The CSV dataset was imported into a relational database (MySQL) to enable structured querying and a live Tableau connection. The following steps were performed:

1. Created a MySQL database schema named heart_disease_db
2. Defined table heart_patients with columns matching all 18 dataset features, with appropriate data types (VARCHAR for categoricals, FLOAT for BMI/sleep, INT for health-day counts)
3. Imported 4,500 records using MySQL LOAD DATA INFILE command
4. Created indexes on frequently filtered columns: HeartDisease, Sex, AgeCategory, Race

3.5 Connecting Database to Tableau

5. Opened Tableau Desktop and selected MySQL as the data source connector
6. Entered connection credentials: Host (localhost), Port (3306), Database (heart_disease_db)
7. Selected the heart_patients table and previewed data in Tableau's data source pane
8. Verified row count (4,500 records) and column types
9. Saved the data connection for use across all worksheets and dashboards

04

DATA PREPARATION

Cleaning, transforming, and preparing the dataset for visualization in Tableau.

4.1 Data Quality Assessment

Before visualization, the dataset was assessed for data quality issues. The following checks were conducted:

- Missing Values: No significant missing values detected. All 4,500 records were complete across all 18 columns.
- Duplicate Records: Checked for exact-row duplicates — none found.
- Outlier Detection: BMI values above 60 were investigated. A small number of extreme values (BMI > 60) were retained as valid obese outlier cases rather than data errors.
- Data Type Validation: All numeric fields (BMI, PhysicalHealth, MentalHealth, SleepTime) confirmed as numeric. Binary fields confirmed as consistent Yes/No strings.

4.2 Data Transformations in Tableau

The following transformations and calculated fields were created within Tableau:

4.2.1 Age Category Grouping

The AgeCategory field was already binned into 5-year intervals. A custom Tableau parameter was created to allow users to group ages into broader bands (Young Adults: 18–34, Middle-Aged: 35–54, Senior: 55+) for high-level filtering.

4.2.2 Calculated Field: HD Rate

Calculated Field: Heart Disease Rate

```
IF [HeartDisease] = 'Yes' THEN 1 ELSE 0 END
// Used as a measure to compute average heart disease prevalence across dimensions
```

4.2.3 Calculated Field: BMI Category

Calculated Field: BMI Category

```
IF [BMI] < 18.5 THEN 'Underweight'
ELSEIF [BMI] < 25 THEN 'Normal'
ELSEIF [BMI] < 30 THEN 'Overweight'
ELSE 'Obese'
END
```

4.3 Data Filters Applied

- Gender Filter: Show/hide Male or Female patients across dashboards
- Age Filter: Slider from 18–80+ to focus on specific age groups
- Race Filter: Multi-select dropdown for racial/ethnic group selection
- General Health Filter: Toggle between health self-ratings (Excellent to Poor)
- Diabetic Status Filter: Select specific diabetic categories for focused analysis

4.4 Data Preparation for Story

For the Tableau Story, data was additionally sorted and ranked so that the narrative flows from high-level demographic insights (gender, age) to deeper clinical factors (diabetes, stroke, BMI), enabling a logical, story-driven analysis progression for non-technical viewers.

05

DATA VISUALIZATIONS

Description of all 10 unique visualizations created across worksheets and dashboards.

5.1 Visualization Summary

A total of 10 unique visualizations were created, spanning 10 individual Tableau worksheets, organized into 3 interactive dashboards and 1 Tableau Story. Each visualization was designed to answer a specific analytical question about heart disease risk factors.

#	Chart Title	Chart Type	Key Insight
1	**Gender vs Heart Disease**	*Bar Chart*	Compares count of heart disease (Yes/No) between Female and Male patients. Females: 2,377 No / 227 Yes. Males: 1,640 No / 256 Yes.
2	**Age vs Heart Disease**	*Grouped Bar Chart*	Stacked bars show heart disease distribution across age categories (18–24 to 80+) segmented by sex. Peak cases observed in 60–80 age range.
3	**Diabetic vs Stroke**	*Bubble/Scatter Chart*	Bubble chart plotting stroke count against diabetes status (No / Borderline / Yes / Yes during pregnancy), split by heart disease status.
4	**Impact of Smoking & Alcohol**	*Horizontal Bar Chart*	4-category breakdown: No-Smoke/No-Alcohol (2,500+), No-Smoke/Alcohol, Smoke/No-Alcohol, Smoke/Alcohol. Shows relative impact on heart disease count.
5	**Stroke vs Heart Disease**	*Multi-panel Bar Chart*	Three-way analysis: Asthma × Kidney Disease × Skin Cancer vs Stroke count, further segmented by heart disease status (Yes/No).
6	**Racewise Heart Disease**	*Pie Chart*	Distribution: White 75.57%, Black 17.81%, Hispanic 3.73%, Asian, American Indian/Alaskan 0.21%, Other races.
7	**General Health vs Heart Disease**	*Bubble/Circle Pack*	Bubble sizes represent patient count per general health rating: Good (172), Fair (140), Poor (92), Very Good (67), Excellent.
8	**Physical Activity vs Heart Disease**	*Donut Chart*	204 physically active vs 279 inactive patients with heart disease. Shows protective effect of regular activity.
9	**Age & BMI vs Diabetic**	*Treemap*	Heatmap treemap of average BMI across age groups and diabetic status. Range: 23.91 (healthy) to 42.78 (obese). Darker = higher BMI.
10	**People Got Stroke — Diabetes & Heart Disease**	*Grouped Bar Chart*	Multi-variable chart: Heart Disease × Diabetic status × Age Category showing average BMI. Identifies compound risk group aged 55–69.

5.2 Detailed Visualization Descriptions

5.2.1 Gender vs Heart Disease (Bar Chart)

Two vertical bars represent Female and Male patient groups. Each bar is color-coded: blue for No heart disease and orange for Yes. The chart immediately reveals that while more female patients (2,377) are present in the dataset, male patients (256) have a slightly higher absolute count of heart disease cases than females (227), despite being a smaller subgroup.

5.2.2 Age vs Heart Disease (Grouped Bar Chart)

Grouped bars across 13 age categories show how heart disease prevalence increases with age. The chart uses Sex as a secondary dimension. Heart disease cases become visually dominant in the 60–80 age range, confirming that age is the strongest demographic predictor in this dataset. Cases below age 40 are minimal.

5.2.3 Diabetic vs Stroke (Bubble/Scatter Chart)

This chart plots stroke count on the Y-axis against four diabetic status categories (No, Borderline, Yes, Yes-during-pregnancy), separated into two panels by heart disease status. The non-diabetic, no-heart-disease group shows the highest stroke avoidance. Diabetic patients with heart disease show significantly elevated stroke counts.

5.2.4 Impact of Smoking and Alcohol on Heart Disease

A horizontal bar chart with 4 rows representing combinations of Smoking (Yes/No) × Alcohol (Yes/No). The longest bar (Non-smoker, Non-drinker) has 2,500+ cases with minimal heart disease. The shortest bars (both lifestyle factors present) show disproportionately higher orange segments, confirming compound lifestyle risks.

5.2.5 Stroke vs Heart Disease (Multi-Panel)

This advanced visualization uses Asthma × Kidney Disease × Skin Cancer as panel dimensions, plotting stroke count per cell. The heat-like pattern reveals that patients with both heart disease and asthma show the highest stroke rates, suggesting a three-way comorbidity relationship.

5.2.6 Racewise Heart Disease (Pie Chart)

A pie chart showing the racial composition of heart disease patients. White: 75.57%, Black: 17.81%, Hispanic: 3.73%, Asian, American Indian/Alaskan Native: 0.21%, Other. This chart must be interpreted in context of dataset demographics rather than assuming inherent racial risk differences.

5.2.7 General Health vs Heart Disease (Bubble Chart)

A bubble/circle pack chart where circle sizes represent patient counts within each self-reported general health category. Good health respondents form the largest bubble (172 HD cases), which seems counterintuitive but reflects the larger population in this category. Poor-rated patients (92) have a much higher HD rate relative to their group size.

5.2.8 Physical Activity vs Heart Disease (Donut Chart)

The donut chart shows 204 physically active vs 279 inactive patients have heart disease. Given that active patients form the majority of the dataset, the normalized rate confirms that physical inactivity carries a significantly higher heart disease risk per capita.

5.2.9 Age & BMI vs Diabetic (Treemap)

A treemap where rectangles represent age-diabetic combinations, sized by count and colored by average BMI (light = ~23.91 to dark teal = ~42.78). The 18–24 Yes-diabetic group shows the highest average BMI, indicating early-onset obesity-driven diabetes. The treemap allows simultaneous comparison across multiple dimensions.

5.2.10 People Got Stroke — Diabetes, Heart Disease & Age

The most complex visualization: a grouped bar chart with Heart Disease × Diabetic status as column headers and Age Category on the X-axis, with Average BMI on the Y-axis. This reveals that the compound risk group (HD = Yes, Diabetic = Yes, Age 55–69) has the highest average BMI and stroke risk, identifying a critical intervention target population.

06

DASHBOARDS

Design, layout, and content of the three interactive Tableau dashboards.

6.1 Dashboard Design Principles

- Consistent color scheme: Blue for No heart disease, Orange/Red for Yes heart disease
- Shared filter pane using Tableau Actions for cross-visual filtering
- Desktop Browser optimized at 1000x800 resolution
- Navigation buttons (Next/Prev) added for guided exploration between dashboards
- Legend panels placed consistently on the right side of each dashboard

6.2 Dashboard 1 — Demographic & Lifestyle Overview

Dashboard 1 contains four visualizations arranged in a 2x2 grid:

- Top-Left: Gender vs Heart Disease (Bar Chart)
- Top-Right: Age vs Heart Disease (Grouped Bar Chart by Age Category/Sex)
- Bottom-Left: Diabetic vs Stroke (Bubble Chart)
- Bottom-Right: Impact of Smoking and Alcohol on Heart Disease (Horizontal Bar)

Key Insight from Dashboard 1: Age 60+ patients, especially those who smoke, represent the highest-risk demographic group. The gender gap narrows significantly in older age brackets.

6.3 Dashboard 2 — Clinical & Activity Analysis

Dashboard 2 focuses on clinical comorbidities and lifestyle quality indicators:

- Top-Left: Stroke vs Heart Disease (Multi-panel — Asthma/Kidney/Skin Cancer)
- Top-Right: Racewise Heart Disease (Pie Chart)
- Bottom-Left: General Health vs Heart Disease (Circle/Bubble Chart)
- Bottom-Right: Physical Activity vs Heart Disease (Donut Chart)

Key Insight from Dashboard 2: Patients rating their health as 'Poor' or 'Fair' have disproportionately high heart disease rates. Physical inactivity is a strong independent predictor.

6.4 Dashboard 3 — BMI, Diabetes & Compound Risk

Dashboard 3 explores the intersection of BMI, diabetes, and complex risk profiles:

- Top: Age and BMI vs Diabetic (Treemap) — spans full width
- Bottom: People Got Stroke — Diabetes, Heart Disease & Age (Grouped Bar Chart)

Key Insight from Dashboard 3: Young diabetic patients (18–24) have alarmingly high average BMI (~42), suggesting early intervention is critical. The compound risk group (HD + Diabetes + Age 55–69) has the highest stroke vulnerability.

6.5 Tableau Public URL

Published Dashboard URL:
https://public.tableau.com/views/heart_disease_dashboard_17728172824120/Dashboard1

07

TABLEAU STORY

A narrated, guided analytical journey through all heart disease visualizations.

7.1 Story Overview

The Tableau Story titled "Heart Disease Analysis Story" organizes all 10 visualizations into a sequenced, narrated presentation that takes the viewer on a logical journey from demographic observations to compound clinical risk factors.

7.2 Story Scenes

Scene	Title	Narrative Focus
1	**Gender vs Heart Disease**	Introduces the dataset. Reveals the gender distribution of heart disease — more females in the dataset but males show proportionally higher risk rates.
2	**Age vs Heart Disease**	Age is the strongest predictor. Cases escalate sharply after age 55. Both genders converge in older brackets.
3	**Diabetic vs Stroke**	Comorbidity analysis: diabetes amplifies stroke risk among heart disease patients significantly.
4	**Impact of Smoking and Alcohol**	Lifestyle choices compound risk. Non-smoker non-drinkers still have the largest group, but relative risk of smokers is 2x+.
5	**Stroke vs Heart Disease**	Three-way comorbidity: Asthma + Kidney Disease + Skin Cancer all correlate with elevated stroke risk in HD patients.
6	**Racewise Heart Disease**	Demographic distribution of heart disease. White population dominates by size; analysis of relative rates important.
7+	**General Health, Physical Activity, BMI & Diabetes**	Synthesizes all prior findings. Paints a picture of the highest-risk individual: older, diabetic, physically inactive, with high BMI.

08

PERFORMANCE TESTING

Validating dashboard performance, data rendering efficiency, and filter responsiveness.

8.1 Performance Test Results

Test Metric	Value	Status	Remarks
Total Records Rendered	**4,500**	**Pass**	All records loaded without timeout
No. of Unique Visualizations	**10**	**Pass**	Across 3 dashboards and 1 story
No. of Dashboards	**3**	**Pass**	Dashboard 1, 2, and 3 fully functional
Story Scenes	**6+**	**Pass**	Full story with narrated scenes per viz
Data Filters Applied	**5+**	**Pass**	Gender, Age, Race, GenHealth, Diabetic filters
Calculated Fields	**3**	**Pass**	Age Category grouping, HD Rate, BMI Category
Dashboard Load Time	**< 5 sec**	**Pass**	Tested on Tableau Public
DB to Tableau Connection	**Active**	**Pass**	MySQL → Tableau live connection verified
Flask Web Integration	**Active**	**Pass**	All dashboards embedded via iframe
Responsive Dashboard Design	**Yes**	**Pass**	Desktop browser optimized (1000x800)

8.2 Performance Observations

- All 4,500 records render successfully in Tableau Public without performance degradation
- Interactive filters respond within 1–2 seconds on standard broadband connections
- The treemap (Dashboard 3) is the most computationally intensive visualization due to multi-dimension aggregation
- Donut chart (Physical Activity) renders fastest due to simple binary dimension
- No data extracts were required — live MySQL connection maintains acceptable performance at this dataset size

09

WEB INTEGRATION

Embedding Tableau dashboards in a Flask web application for public accessibility.

9.1 Technology Stack

- Backend: Python Flask (v3.0) — routing, API endpoints, CSV data serving
- Frontend: HTML5, CSS3, JavaScript — responsive single-page styled website
- Tableau Embedding: Tableau Public iframe embed with tab-based navigation
- Charts: Chart.js for dataset overview charts (gender, HD prevalence, general health, lifestyle)
- Dataset: Heart_new2.csv served directly via Flask /api/stats and /api/sample endpoints

9.2 Website Pages

- Home Page (/): Hero section, live statistics from CSV, project scenarios, features grid, tech stack
- Dashboard Page (/dashboard): Tabbed Tableau embed for Dashboard 1, 2, 3, and Story with insight cards
- Dataset Page (/dataset): All 18 column descriptions, 4 Chart.js charts, sample data table (first 20 rows)
- Team Page (/team): Team lead and member cards, SmartBridge program info, project statistics

9.3 Tableau Embed Code Approach

Tableau Iframe Embed (dashboard.html)

```
<iframe
  src='https://public.tableau.com/views/heart_disease_dashboard_17728172824120/Dashboard1
    ?:embed=yes&:showVizHome=no&:tabs=yes&:toolbar=yes'
  width='100%' height='900' frameborder='0'>
</iframe>
// JavaScript tab switching dynamically updates the iframe src for each dashboard
```

9.4 Flask API Endpoints

Endpoint	Method	Description
GET /	GET	Renders homepage with hero, stats, scenarios, and feature sections
GET /dashboard	GET	Renders the Tableau dashboard embed page with tab navigation
GET /dataset	GET	Renders dataset explorer with column info, charts, and sample table
GET /team	GET	Renders team members page with roles and SmartBridge program stats
GET /api/stats	GET (JSON)	Returns computed stats from CSV: total, HD rate, smokers, stroke, gender breakdown, race, general health
GET /api/sample	GET (JSON)	Returns first 20 rows of the heart disease CSV as JSON for table display

10

RESULTS & KEY FINDINGS

Analytical conclusions derived from the Tableau dashboard visualizations.

10.1 Summary of Results

Heart Disease Prevalence in Dataset

Total Patients: 4,500 | Heart Disease Cases: 483 (10.73%)
Female Patients — No HD: 2,377 | Yes HD: 227
Male Patients — No HD: 1,640 | Yes HD: 256
Most Affected Age Group: 60–80 years across both genders
Dominant Race in Dataset: White (75.57%), Black (17.81%), Hispanic (3.73%)

10.2 Key Analytical Findings

Finding 1: Age is the Dominant Risk Factor

Heart disease cases increase exponentially after age 55, with the 65–74 bracket showing the highest absolute count. Age-targeted preventive programs should be initiated by age 45 for early intervention.

Finding 2: Diabetes Significantly Amplifies Risk

Diabetic patients (any status) show markedly higher stroke counts when combined with heart disease. The compound risk of diabetes + heart disease + age 55+ is the most dangerous patient profile in the dataset.

Finding 3: Physical Inactivity is a Strong Predictor

Despite active patients forming the larger group, inactive patients (279) had more heart disease cases than active patients (204). The per-capita heart disease rate in inactive patients is estimated at 2x that of active patients.

Finding 4: BMI and Obesity Drive Early-Onset Risk

Young diabetic patients (18–24) show the highest average BMI (~42.78), suggesting obesity-driven diabetes in younger populations. This age group requires early nutritional and lifestyle intervention.

Finding 5: Smoking Compounds, But Isn't Dominant Alone

Non-smokers, non-drinkers form the largest heart disease group simply due to their dominant representation in the dataset. However, when normalized by group size, smokers show ~2x higher relative risk, confirming that smoking cessation is a high-priority public health goal.

Finding 6: Self-Reported Health Correlates with Diagnosis

Patients rating their health 'Poor' (92 HD cases) have a much higher per-capita rate than those rating themselves 'Excellent'. The General Health dimension serves as a reliable screening proxy, suggesting that simple self-assessment questionnaires can be effective early-warning tools.

10.3 Recommendations

- Healthcare providers should prioritize cardiology screening for patients aged 55+ with diabetes and high BMI
- Public health campaigns should focus on smoking cessation and physical activity promotion in middle-aged demographics
- Government policies should incentivize physical activity infrastructure and healthy food access in lower-income communities
- Hospital intake forms should routinely capture lifestyle indicators (smoking, activity, BMI) to enable proactive risk stratification
- Tableau dashboards should be made available to ward-level clinical staff to enable data-driven patient triaging

11

PROJECT DOCUMENTATION & CONCLUSION

Step-by-step development log, team contributions, and project conclusion.

11.1 Step-by-Step Project Development Log

Step	Phase	Activities Completed	Responsible
1	**Problem Definition**	Defined business problem, identified stakeholders (Dr. Sharma, Ramesh, Anita), wrote business requirements	Anagh (Lead)
2	**Literature Survey**	Reviewed WHO CVD statistics, CDC BRFSS documentation, Tableau healthcare case studies	All Members
3	**Data Collection**	Obtained Heart Disease CSV dataset; validated 4,500 records, 18 features	Rohit Patil
4	**Database Setup**	Created MySQL DB schema heart_disease_db, defined table with appropriate types, imported CSV	Rohit Patil
5	**SQL Operations**	Wrote exploratory SQL queries for gender counts, BMI averages, race distribution, diabetic analysis	Rohit Patil
6	**DB-Tableau Connect**	Connected MySQL database to Tableau Desktop via MySQL connector, verified data integrity	Rohit Patil + Anagh
7	**Data Preparation**	Cleaned data, created calculated fields (HD Rate, BMI Category, Age Groups), applied filters	Prapti These
8	**Visualizations (1–5)**	Built first 5 worksheets: Gender, Age, Diabetic/Stroke, Smoking/Alcohol, Stroke/HD	Anagh + Prapti
9	**Visualizations (6–10)**	Built remaining 5: Racewise, General Health, Physical Activity, BMI Treemap, Compound Risk	Rutuj Bhandari
10	**Dashboard 1–3**	Assembled three dashboards, applied tiled layout, added legends, navigation buttons, filters	Anagh + Rutuj
11	**Tableau Story**	Created 7-scene story with narrated captions published on Tableau Public	Anagh Navarkar

12	**Performance Testing**	Verified data rendering, filter responsiveness, calculated field count, load time	Rutuj Bhandari
13	**Web Integration**	Built Flask app, HTML/CSS/JS pages, embedded Tableau, deployed Chart.js dataset charts	Shlok Bam
14	**Documentation**	Wrote complete project documentation per SmartBridge template	Shlok Bam + All
15	**Review & Submission**	Team lead review, final corrections, Tableau Public publish, GitHub and demo links	Anagh Navarkar

11.2 Team Contributions

Name	Role	Responsibilities
Anagh Navarkar	**Team Lead**	Project coordination, dashboard design oversight, story creation, final review and submission management
Prapti These	*Member*	Data preparation, cleaning, and transformation; visualization design for Dashboard 2
Rohit Patil	*Member*	Database setup, SQL operations, data extraction, Tableau-DB connection configuration
Rutuj Bhandari	*Member*	Dashboard 3 development, performance testing, treemap and multi-variable chart creation
Shlok Bam	*Member*	Flask web integration, HTML/CSS/JS development, Tableau embedding, project website

11.3 Conclusion

This project successfully demonstrates the power of data visualization as a tool for extracting meaningful healthcare insights. Using Tableau as the primary BI platform, the team built 10 unique interactive visualizations organized into 3 dashboards and a narrated story, analyzing heart disease data across dimensions of demographics, lifestyle, and clinical comorbidities.

The key finding — that age, diabetes, physical inactivity, and high BMI collectively define the highest-risk patient profile — is communicated clearly through Tableau's interactive and visual medium, making it actionable for cardiologists, policy makers, and patients alike.

The Flask web integration ensures that these insights are accessible beyond Tableau Desktop users, providing a public-facing URL for broader stakeholder engagement. The project meets all SmartBridge deliverable requirements across data collection, preparation, visualization, dashboard design, story creation, performance testing, and web integration.

11.4 Future Enhancements

- Incorporate machine learning predictions (logistic regression) to score individual patient heart disease risk probability
- Connect to a live hospital data feed for real-time dashboard updates
- Build a patient-facing mobile application that generates personalized risk reports
- Expand dataset to include blood pressure, cholesterol levels, and medication history for deeper clinical analysis
- Add geospatial analysis layer to compare regional/state-level heart disease prevalence

11.5 References

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End of Document — Heart Disease Analysis | Smart Bridge Data Analytics with Tableau